PROJECT SENTINEL-BLR

Evidence-Quality Automated Traffic Enforcement Framework

Submission Details	
Hackathon	Flipkart Gridlock 2.0
Phase	Phase 2
Theme	Theme 3 — Automated Photo Identification and Classification for Traffic Violations
Submission Type	Framework & Prototype Proposal
Version	v6.0 — Legal & Cost Validation Pass; Corrected MV Act Citations; DPDP Act Compliance
Submitted By	Team_Rocket

Summary at a Glance

The idea: Sentinel-BLR doesn't just detect violations — it scores how *reliable* each detection is before deciding whether to auto-fine, human-review, or discard it. That score is the **Evidence Reliability Score (ERS)**, a single 0–100 number combining detection confidence, OCR confidence, multi-frame consensus, calibration health, and image quality.

$$\mathrm{ERS} = 0.30\, C_{\mathrm{det}} + 0.25\, C_{\mathrm{ocr}} + 0.20\, R_{\mathrm{consensus}} + 0.15\, H_{\mathrm{cal}} + 0.10\, Q_{\mathrm{pre}}$$

Why it matters: ERS $\geq 85 \rightarrow$ auto-fine. 60–84 $\rightarrow$ human review. $< 60 \rightarrow$ discard, no record. One number replaces five fields for officers and courts — and structurally prevents a fine being issued on weak evidence.

Three structural fixes most pipelines skip:

- **Camera drift** — self-healing Auto-Calibration Heartbeat (SIFT/ORB homography) every 15 min, no enforcement downtime.
- **Unreadable plates** — 5-frame character-level probability fusion recovers OCR even from mud/rain-damaged plates.
- **Policy as YAML, not model weights** — BTP can change enforcement rules in <1 minute without engineer involvement.

Cost (Phase 1, 10 junctions): Edge hardware alone runs in the low lakhs — Jetson Orin NX modules at ~\$600–970/unit, reusing existing BTP CCTV feeds rather than replacing them. See Section 7 onward and the Cost and Legal Feasibility section for the full breakdown.

Legal grounding (verified, not assumed): §129 (helmet), §128/§194C (triple riding), §184(e) (wrong-side — corrected from an earlier draft's §185 miscitation), §194D (repeat-offence

escalation), and explicit Digital Personal Data Protection Act 2023 alignment (face-blurring, 24-hour raw-frame deletion). Earlier citation errors were caught and corrected in this version — see Section 15.

Deployment path: 10 junctions → 50 junctions → city-wide, each gated by a measurable go/no-go criterion, not a timeline guess.

Not a greenfield problem: Bengaluru's own ITMS, Hyderabad's ANPR e-challan system, and Delhi's RLVD/OSVD network already detect violations at scale. None of them, on current public evidence, score whether a detection is reliable enough to enforce on — and that gap produced a real 26,000-challan failure in Hyderabad in 2025. See Section 4.

What's proven vs. what's a design target: OCR fusion, the calibration heartbeat, and the policy-engine pattern rest on established technique. The ≤28 ms standard-condition compute budget and the Volumetric Bounding-Box triple-riding threshold are explicit design targets pending Phase 1 field validation — and Layer 4A already structurally prevents the least-validated component (triple riding) from ever auto-fining, regardless of confidence. Sentinel-BLR is built to know what it doesn't know yet, not to overclaim certainty it hasn't earned.

Synthetic priors — data source defined: Safety-violation count initialisers are absent from the provided datasets (2 helmet and 0 triple-riding records across 298,450 rows in the Police Violations Dataset). Layer 8 initialises violation-type weights to uniform and relies on pre-trained IDD/ACFR/Roboflow Helmet weights as the classification foundation. These priors carry <20% weight from day 1 and are replaced entirely by live field observations within 30 days of Phase 1 operation. They are never used for urgency scoring or auto-fine logic.

1. Executive Summary

Sentinel-BLR is not a traffic violation detector. It is an evidence-quality system — one that determines when a detected violation is legally reliable enough for enforcement.

Most submissions for this problem will bolt together YOLO, OCR, and a violation classifier. Sentinel-BLR does that too — and then solves the three problems that make those pipelines fail in actual Bengaluru deployment:

- **Cameras that drift** — addressed by a self-healing Auto-Calibration Heartbeat (SIFT/ORB + inverse homography H^{-1}) detecting and correcting spatial drift every 15 minutes, without halting enforcement.
- **Plates that are unreadable** — addressed by Temporal Character-Level Probability Fusion across 5 frames with Levenshtein outlier filtering, producing a majority-vote plate string even from mud-damaged plates.
- **Detection confidence that does not translate into legally defensible enforcement** — addressed by per-violation-type confidence thresholds, a configuration-file policy engine (not an ML model), and a composite Evidence Reliability Score (ERS) on every evidence packet.

Sentinel-BLR processes live camera feeds under real-world environmental stress — night, rain, motion blur, monsoon fog, sodium vapour streetlighting — and detects all seven categories of traffic violation defined in the problem statement.

Four Architectural Principles

1. Detect violations reliably under adverse Bengaluru conditions.
2. Gate enforcement on evidence quality, not raw detection confidence.

3. Keep all enforcement policy inspectable and updateable without model redeployment.
4. Produce legally defensible, tamper-evident evidence packets on every confirmed violation.

2. Evidence Reliability Score — The Centrepiece Innovation

Every other traffic enforcement pipeline asks a single question: *was a violation detected?* Sentinel-BLR asks a second question that no current Indian deployment asks: *is the evidence reliable enough to stand up in enforcement and in court?*

The answer is a single composite integer on every evidence packet — the **Evidence Reliability Score (ERS)**. No new model is required; all five input components are computed upstream as part of the standard pipeline.

ERS Formula

$$\mathrm{ERS} = \underbrace{0.30 \times C_{\mathrm{det}}}_{\text{detection}} + \underbrace{0.25 \times C_{\mathrm{ocr}}}_{\text{plate read}} + \underbrace{0.20 \times R_{\mathrm{consensus}}}_{\text{multi-frame}} + \underbrace{0.15 \times H_{\mathrm{cal}}}_{\text{calibration}} + \underbrace{0.10 \times Q_{\mathrm{pre}}}_{\text{image quality}} \tag{1}$$

All five components are computed upstream as part of the standard pipeline; the ERS introduces no additional inference overhead.

ERS Weight Rationale

The weights are not arbitrary — they are derived from a failure-consequence hierarchy specific to automated enforcement. Each component is weighted by the severity of the error it guards against:

Component	Weight	Rationale
Detection confidence (C_{det})	0.30	Highest weight: a misclassified violation type (e.g. legal manoeuvre flagged as wrong-side) produces a wrongful fine. No downstream check recovers from a wrong detection class.
OCR confidence (C_{ocr})	0.25	Second highest: a misread plate issues a fine to the wrong registered owner — a legal liability and a public trust failure. Plate identity is the sole link between evidence and enforcement action.
Frame consensus ($R_{\mathrm{consensus}}$)	0.20	A single-frame detection may be a transient artefact (shadow, reflection). Multi-frame agreement is the primary mechanism for suppressing spurious detections; weighted accordingly.
Calibration health (H_{cal})	0.15	A drifted camera silently shifts spatial ROIs, producing systematic false positives at stop-lines and parking zones. Important, but recoverable: the heartbeat flags drift and re-projects automatically.
Preprocessing quality (Q_{pre})	0.10	Lowest weight: image degradation affects all upstream components and is already reflected in their confidence scores. This term captures residual uncertainty not absorbed elsewhere (e.g. extreme sodium-vapour saturation).

The top two components together account for 55% of ERS, directly reflecting that vehicle misidentification and plate misread are the two failure modes most likely to result in a wrongful fine or a successful legal challenge to an enforcement action.

ERS Weight Calibration Status — What Is Claimed and What Is Not

The weights (0.30, 0.25, 0.20, 0.15, 0.10) are derived from a **failure-consequence hierarchy**, not from empirical optimisation on labelled enforcement outcomes. This distinction is stated here deliberately: the hierarchy is logically well-motivated (a misidentified vehicle type cannot be recovered by any downstream check; a misread plate issues a fine to the wrong owner; a single-frame transient differs from a multi-frame event), but the precise numerical values are Phase 1 calibration targets, not validated parameters.

Known correlation between components. Q_{pre} is not fully independent of C_{det} and C_{ocr}: severe image degradation depresses all three simultaneously. In practice, the linear formula structurally *amplifies* the penalty for bad-image events — a low Q_{pre} frame also yields a lower C_{det} and C_{ocr}, so the effective ERS penalty from image degradation is larger than the 0.10 weight alone suggests. Section 2 below quantifies this amplification on a concrete heavy-rain example and explains why it produces a desirable overcaution property rather than a miscalibration.

Phase 1 calibration plan. Every human accept/reject decision in the 60–92% confidence band generates a labelled outcome. After 500 labelled decisions, a constrained least-squares fit will re-estimate weights subject to the constraint $\sum w_i = 1$, $w_i > 0$. Because Q_{pre} is partially collinear with C_{det} and C_{ocr} (above), the fit is *ridge-regularised* (an L2 penalty on the weight vector): an unregularised least-squares fit on correlated regressors yields unstable, high-variance coefficients, so the regulariser is what keeps the re-estimated weights interpretable rather than an artefact of which collinear term the optimiser happened to load. If the fitted weights diverge from the initial hierarchy by more than ± 0.08 on any component, the ERS formula will be updated and the routing thresholds re-validated before Phase 2 deployment. The initial weights are the prior; live enforcement data provides the posterior.

Mock Evidence Packet — What an Enforcement Officer Receives

The table below illustrates a representative evidence packet output for a helmet non-compliance detection, demonstrating how the ERS translates raw model outputs into a single, actionable enforcement decision:

Field	Value
Violation Type	Helmet Non-Compliance
Vehicle Class	2-Wheeler
Plate Text	KA 05 MN 4821
OCR Confidence	94% (3/5 frames consensus; Levenshtein-clean)
Detection Confidence	91% (3-frame consensus; MobileNetV3)
Calibration Health	HEALTHY — last heartbeat 8 min ago, drift 0.3px
Frame Consensus	5/5 frames in agreement
Preprocessing Quality	CLAHE + Retinex active (low-light condition)
SHA-256 Hash	`3fa2...c91b` (frame tamper-evident)
Evidence Reliability Score	**ERS = 91 / 100 — AUTO-FINE ELIGIBLE**

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Field	Value
(continued)	
MV Act Reference	§129 (helmet); §194D enhanced penalty on subsequent offence if repeat offender flagged

ERS Routing Logic

ERS Range	Decision	Rationale
85 – 100	**AUTO-FINE**	All components strong; legally defensible without human review
60 – 84	**HUMAN REVIEW**	One or more components borderline; officer adjudicates
< 60	**DISCARD**	Evidence quality insufficient; no fine, no record

ERS in Practice — Three Worked Scenarios and Boundary Sensitivity

The following three scenarios compute ERS end-to-end on representative operating conditions. Component values reflect realistic model outputs under those conditions; the computations demonstrate that ERS routing is fully deterministic and reproducible from five upstream measurements with no additional judgement.

Component (weight)	Scenario A	Scenario B	Scenario C
Condition			
	Clear day, CLAHE active only	Light rain, 3/5 frames usable	Heavy rain + damaged plate
C_{det} — detection ($\times 0.30$)	91	81	62
C_{ocr} — plate read ($\times 0.25$)	94	68	28
$R_{\text{consensus}}$ — multi-frame ($\times 0.20$)	95	75	40
H_{cal} — calibration ($\times 0.15$)	100	88	70
Q_{pre} — image quality ($\times 0.10$)	85	55	35
ERS (computed)	**93.3**	**75.0**	**47.6**
Routing decision	**AUTO-FINE**	**HUMAN REVIEW**	**DISCARD**

Scenario A (clear day, helmet non-compliance):

$$\begin{aligned}\text{ERS} &= (0.30 \times 91) + (0.25 \times 94) + (0.20 \times 95) + (0.15 \times 100) + (0.10 \times 85) \\ &= 27.3 + 23.5 + 19.0 + 15.0 + 8.5 \;=\; \mathbf{93.3} \;\rightarrow\; \text{AUTO-FINE}\end{aligned}$$

Scenario B (light rain, partial plate occlusion):

$$\begin{aligned}\text{ERS} &= (0.30 \times 81) + (0.25 \times 68) + (0.20 \times 75) + (0.15 \times 88) + (0.10 \times 55) \\ &= 24.3 + 17.0 + 15.0 + 13.2 + 5.5 \;=\; \mathbf{75.0} \;\rightarrow\; \text{HUMAN REVIEW}\end{aligned}$$

Scenario C (heavy rain, mud-damaged plate):

$$\begin{aligned}\text{ERS} &= (0.30 \times 62) + (0.25 \times 28) + (0.20 \times 40) + (0.15 \times 70) + (0.10 \times 35) \\ &= 18.6 + 7.0 + 8.0 + 10.5 + 3.5 \;=\; \mathbf{47.6} \;\rightarrow\; \text{DISCARD}\end{aligned}$$

Boundary Sensitivity — What Changes at ERS = 84 vs. 85?

The boundary between AUTO-FINE and HUMAN REVIEW is the decision point with the largest enforcement consequence. The scenario below constructs a case sitting just below the 85 threshold and analyses what would happen under two alternative policy choices.

Boundary scenario (marginal detection, calibration drift approaching limit):

- $C_{\text{det}} = 88$ (detection confident, 3-frame consensus met)
- $C_{\text{ocr}} = 91$ (plate read confirmed, Levenshtein-clean)
- $R_{\text{consensus}} = 80$ (3/5 frames, not 5/5 — two frames motion-blurred)
- $H_{\text{cal}} = 78$ (calibration heartbeat 13 min ago, drift 1.1 px — approaching the 1.2 px alert threshold)
- $Q_{\text{pre}} = 65$ (CLAHE + Retinex active, moderate sodium-vapour distortion)

$$\begin{aligned}\text{ERS} &= (0.30 \times 88) + (0.25 \times 91) + (0.20 \times 80) + (0.15 \times 78) + (0.10 \times 65) \\ &= 26.4 + 22.75 + 16.0 + 11.7 + 6.5 \;=\; \mathbf{83.35} \;\rightarrow\; \text{HUMAN REVIEW}\checkmark\end{aligned}$$

Policy variant	ERS	Decision	Risk implication
Baseline (threshold = 85)	83.35	**REVIEW**	Officer adjudicates; no wrongful fine
Threshold lowered to 80	83.35	**AUTO-FINE**	Camera drift at 1.1 px + motion blur = exactly Hyderabad's reported sync-failure mode; auto-fine would issue

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Policy variant	ERS	Decision	Risk implication
(continued)			
OCR weight shifted to 0.20 (redistributed to $R_{\text{consensus}}$ 0.25)	82.8	REVIEW	Same decision; delta = −0.55. This packet sits 1.65 below the gate, so a ±0.05 weight shift (≈ ±0.55 ERS) cannot flip it. Stability exactly at the 84/85 boundary is a Phase 1 calibration check, not claimed from this case
OCR weight shifted to 0.30 (redistributed from $R_{\text{consensus}}$ 0.20 to 0.15)	83.9	REVIEW	Delta = +0.55; still REVIEW

Why the 85 threshold holds for this case. $H_{\text{cal}} = 78$ signals a camera approaching its drift alert boundary. A routing policy that auto-fined this packet would replicate the failure mode documented in Hyderabad's 26,000-challan episode: a weak individual signal (H_{cal} borderline, $R_{\text{consensus}}$ at 80 not 95) passing through a threshold set too low. The 85 boundary ensures that *no single strong component can carry a weak packet to auto-fine*: even with $C_{\text{det}} = 88$ and $C_{\text{ocr}} = 91$, the borderline calibration and incomplete consensus pull ERS below the gate.

The Q_{pre} Correlation Effect — A Feature, Not a Flaw

In Scenario C ($Q_{\text{pre}} = 35$), severe image degradation does not merely subtract $0.10 \times (85 - 35) = 5.0$ points from ERS. It simultaneously depresses C_{det} by approximately 29 points and C_{ocr} by approximately 66 points relative to clear-condition baselines. The total ERS impact of image degradation in Scenario C:

$$
\begin{aligned}
\text{Direct } Q_{\text{pre}} \text{ penalty:} \quad & 0.10 \times (85 - 35) = 5.0 \text{ points} \\
\text{Cascaded via } C_{\text{det}}: \quad & 0.30 \times (91 - 62) = 8.7 \text{ points} \\
\text{Cascaded via } C_{\text{ocr}}: \quad & 0.25 \times (94 - 28) = 16.5 \text{ points} \\
\textbf{Total degradation impact:} \quad & \mathbf{30.2} \text{ points vs. } 5.0 \text{ from } Q_{\text{pre}} \text{ alone}
\end{aligned}
$$

This amplification means any severe-image-quality event drives ERS well below 60 regardless of what the detection model reports — a structurally desirable property. Mud on a lens, heavy monsoon rain, or extreme sodium-vapour saturation produces DISCARD outcomes rather than borderline HUMAN REVIEW outcomes that might produce inconsistent officer adjudication. Phase 1 will measure whether this amplification is appropriately calibrated or creates an overcaution bias in specific lighting conditions (e.g. moderate rain that degrades OCR but leaves detection reliable).

3. System at a Glance — Naive Approach vs. Sentinel-BLR

The pipeline comparison below illustrates, at each processing stage, the structural gap between a standard YOLO+OCR approach and the Sentinel-BLR evidence-quality framework. Each layer on the right directly addresses a documented failure mode of the left.

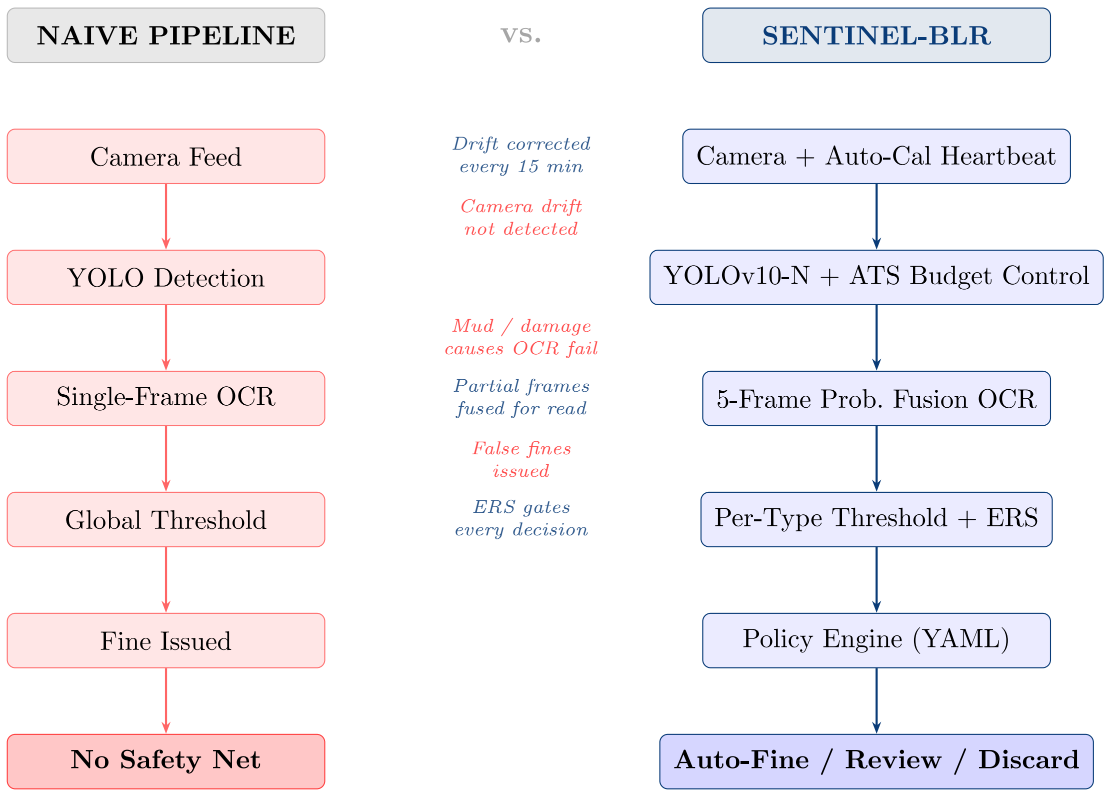

Figure 1: Standard YOLO+OCR pipeline (left) vs. Sentinel-BLR evidence-quality pipeline (right). Red annotations mark naive failure modes; blue annotations indicate the corresponding Sentinel-BLR mitigation.

Problem	**Naive Approach**	**Sentinel-BLR v6.0**
Camera drift	Manual recalibration	Auto-Calibration Heartbeat (SIFT/ORB + H^{-1} ROI re-projection) every 15 min
Dirty / occluded plates	Single-frame OCR	Temporal Character-Level Probability Fusion across 5 frames + Levenshtein outlier filter
Dense traffic triple riding	Pose estimation (fails in crowds)	Volumetric Bounding-Box Profiling (primary, crowd index > 0.35); MediaPipe Pose Lite (enhancement, index ≤ 0.35); 3-frame consensus
False automated fines	Global confidence threshold	Per-violation thresholds + Layer 4A YAML policy engine + ERS; OCR confirmation required
Compute feasibility	Theoretical benchmark	ATS + TensorRT INT8 $\rightarrow \leq$28 ms standard / $\leq$38 ms adverse-peak design target on Jetson Orin NX (see compute budget note in Section 7)

Problem	Naive Approach	Sentinel-BLR v6.0
Kafka bandwidth	Raw frame streaming	Hybrid Evidence Payload: JSON + ROI JPEG crop (<50 KB)
Safety violation data scarcity	Zero-initialise or ignore	Uniform priors (safety violations are absent from provided datasets) + pre-trained IDD/ACFR/Roboflow Helmet weights as classification foundation; live data dominates after 30 days

4. Competitive Landscape — What Already Runs on Bengaluru's Roads

This is not a greenfield problem, and treating it as one would undersell what Sentinel-BLR actually contributes. Bengaluru already operates an AI-based enforcement system, and at least two other major Indian metros run comparable infrastructure at larger scale. The naive pipeline in Section 3 is a pedagogical baseline, not a claim that no automated system exists today — the comparison that actually matters is against what BTP and peer cities have deployed, what they say it does, and what has gone wrong with it in practice.

What's Already Deployed

City / System	Stated Capability	Documented Gap
Bengaluru — ITMS (live since Dec 2022)	250 ANPR cameras + 80 red-light violation cameras across 50 junctions; stated to detect speeding, red-light/stop-lane, helmetless riding, triple riding, mobile-phone use, and seatbelt non-compliance [5]	BTP's own Joint Commissioner of Police (Traffic) stated in 2024 that ITMS does not detect violations including wrong-side driving and footpath riding, and that registering such cases from CCTV footage "is a manual process" — prompting a renewed physical-enforcement drive for exactly the violation categories this problem statement lists [6]
Hyderabad / Cyberabad — ANPR e-challan (since 2018)	City-wide ANPR plus violation-classifier pipeline issuing automated challans for overspeeding, signal jumping, helmet, and seatbelt offences [7]	In the first ten months of 2025, the Cyberabad, Hyderabad, and Rachakonda commissionerates together issued over 26,000 wrong challans. Officials attributed the bulk of these to the violation-detection system and the plate-reading system falling out of sync with each other, including challans issued to vehicles in unrelated parts of the city [8]

City / System	Stated Capability	Documented Gap
Delhi — ITMS / RLVD–OSVD	334 radar-based Red-Light Violation Detection and Over-Speed Violation Detection cameras installed, with 328 more under procurement, as part of an AI-based ITMS monitoring movement, violations, and accidents [9]	Public reporting on the programme centres on camera counts and procurement timelines, not on plate-OCR accuracy, evidence-reliability scoring, or auto-fine precision under adverse conditions — the specific questions this problem statement and Sentinel-BLR are built to answer

Why This Reframes Sentinel-BLR's Contribution

Two conclusions follow directly from the table above, and both sharpen rather than weaken the pitch.

First, Bengaluru does not need another violation detector — it needs the layer its own system is missing. BTP has already stated, on record, which violation categories its existing AI infrastructure cannot yet automate safely: wrong-side driving and footpath/lane violations, both disproportionately exposed to false positives, where a wrong call is a wrongful fine rather than merely a missed one. Sentinel-BLR is not positioned as a replacement for ITMS's 250 ANPR and 80 RLVD cameras. It is positioned as the evidence-reliability layer — ERS (Section 2), per-violation confidence gating, and the Layer 4A policy engine — that would let BTP extend automated enforcement into precisely the categories it has publicly said it cannot yet trust to a machine, without having to re-architect the camera network it has already paid for.

Second, Hyderabad's 2025 wrong-challan episode is not a hypothetical risk invented for this proposal — it already happened, at comparable scale, on a comparable architecture. A detection model and an OCR model running independently and silently drifting out of sync is exactly the failure mode that issues a fine to the wrong vehicle. This is precisely what Layer 5's temporal character-level probability fusion and the ERS's 0.25 weight on OCR confidence exist to prevent: no auto-fine is issued unless plate identity *and* violation detection are jointly confident, not separately plausible and independently wrong. Where Hyderabad's reported failure mode comes from treating detection and OCR as two systems that happen to agree, Sentinel-BLR treats their joint reliability as the enforcement decision itself.

The honest claim is therefore not "no one has built AI traffic enforcement in Indian cities before." It is: **deployed systems already detect violations; none of them, on current public evidence, structurally score whether that detection is reliable enough to enforce on** — and the absence of that layer has already produced a real, reported, 26,000-challan failure in a comparable city deployment. Sentinel-BLR's central innovation exists for a failure mode that has already occurred elsewhere, not a theoretical one.

5. System Architecture — Full Pipeline v6.0

The pipeline processes camera feeds through fifteen distinct layers, each with a defined responsibility and explicit computational budget. Detection and enforcement are deliberately separated: Layer 3 detects; Layer 4A applies enforcement policy as a configuration file, not an ML model.

Three Evidence-Quality Layers — Layer 1A (auto-calibration), Layer 2A (inference budget controller + ATS), and Layer 4A (enforcement policy engine) — together constitute the architecture's spine.

Pipeline Architecture Diagram

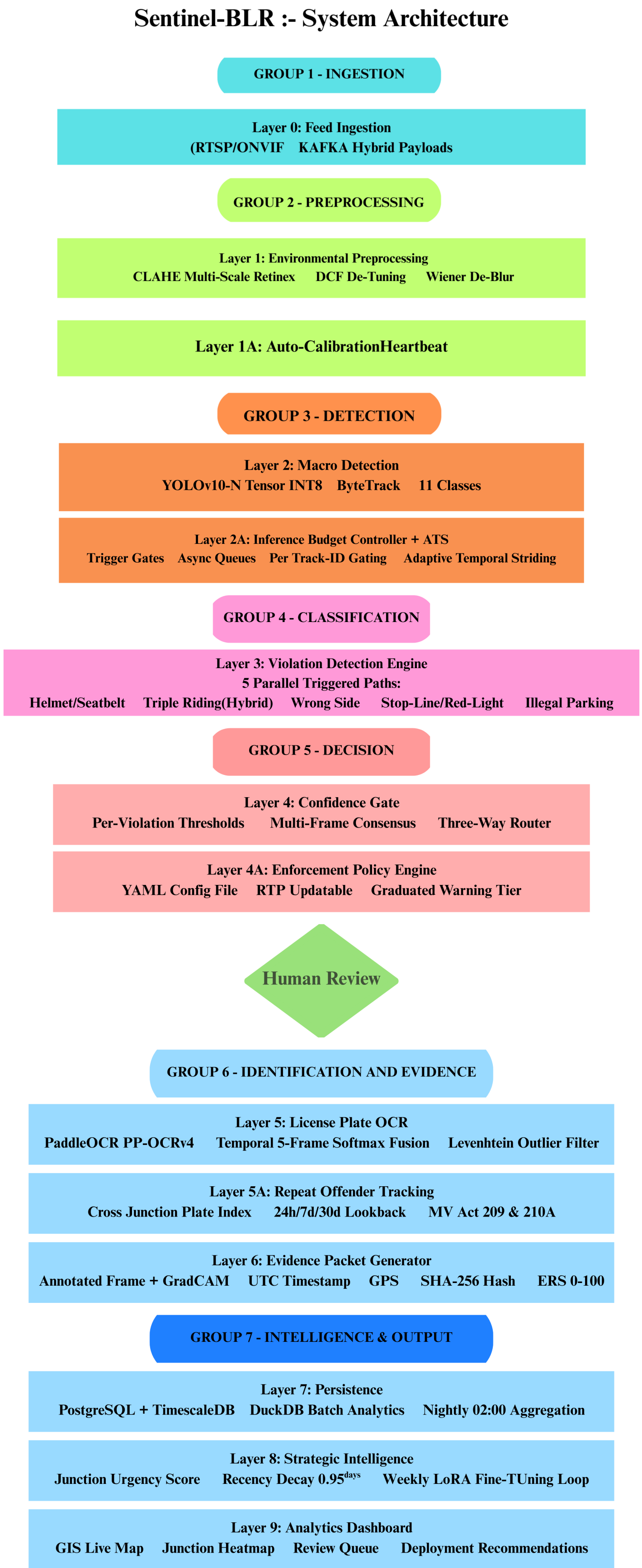

Figure 2: Sentinel-BLR Full System Architecture

Layer Reference Table

Layer	Function	Technical Detail	Group
0	Feed Ingestion	RTSP/ONVIF camera streams; edge-local ring buffer; Kafka Hybrid Evidence Payloads (JSON + ROI JPEG); dynamic 5–30 fps per junction	INGESTION
1	Environmental Preprocessing	CLAHE, Multi-Scale Retinex, DCP de-hazing, Wiener de-blur; GPU-accelerated; conditionally activated per frame	PREPROC
1A ★	Auto-Calibration Heartbeat	SIFT/ORB landmark extraction every 15 min; homography H; drift alert > 1.2 px; inverse H^{-1} re-projects all spatial ROIs automatically	PREPROC
2	Macro Detection	YOLOv10-N TensorRT INT8 at 640px (∼4–7 ms); ByteTrack edge-local; 11 classes	DETECTION
2A ★	Inference Budget Controller + ATS	Trigger-gated sub-model activation; async priority queues; per-Track-ID gating (0.5s window); Adaptive Temporal Striding (every 5th frame in adverse weather)	DETECTION
3	Violation Detection Engine	5 parallel triggered paths: (A) Helmet/Seatbelt, (B) Triple Riding hybrid, (C) Wrong-Side, (D) Stop-Line/Red-Light, (E) Illegal Parking	CLASSIF.
4	Confidence Gate	Per-violation-type multi-frame consensus thresholds; three-way escalation router: AUTO-FINE / HUMAN REVIEW / DISCARD	DECISION
4A ★	Enforcement Policy Engine	YAML config file (not ML); BTP-updatable in <1 min; graduated warning tier; diversion polygon support; OCR-gate and calibration-gate enforcement rules	DECISION
5	License Plate OCR	PaddleOCR PP-OCRv4; Temporal Character-Level Probability Fusion (5-frame softmax sum + argmax); Levenshtein outlier filter; KA-regex gate	ID
5A	Repeat Offender Tracking	Cross-junction plate-violation index; 24h/7d/30d lookback; REPEAT OFFENDER flag; enhanced subsequent-offence penalties under MV Act 2019 §194C/§194D; HIGH CONFIDENCE writes only	ID

Layer	Function	Technical Detail	Group
6A	GradCAM Explainability	Saliency overlay on all flagged frames; async low-priority queue (no pipeline latency); legal defensibility and active learning	EVIDENCE
6	Evidence Packet Generator	Annotated frame + GradCAM + UTC + GPS + junction ID + plate + violation type + vehicle class + SHA-256 + **ERS (0–100)**	EVIDENCE
7	Persistence	PostgreSQL + TimescaleDB; DuckDB batch analytics; nightly 02:00 aggregation	STORAGE
8	Strategic Intelligence	Junction Urgency Score; recency-decay 0.95^{days}; synthetic priors <20% weight after 30 days; weekly LoRA fine-tuning	INTEL
9	Analytics Dashboard	GIS Live Map; Junction Heatmap; Peak-Hour Trends; Human Review Queue; Deployment Recommendations; Model Performance; Calibration Health; Public Compliance Feed	OUTPUT

6. The Six Key Differentiators

The six properties below are what separate Sentinel-BLR from a standard YOLO+OCR assembly. Each is a direct response to a documented failure mode; full technical specification appears in Section 7. Three of the six (character-level OCR fusion, the calibration heartbeat, and policy-as-config) rest on established technique. The remaining three (ATS, the compute budget, and the triple-riding hybrid) are explicit design targets gated behind Phase 1 field validation — and the policy engine in Layer 4A is deliberately structured so that the least-validated component, triple riding, can never auto-fine regardless of confidence. The system is designed to be honest about its own uncertainty rather than to assume it away.

1. **Adaptive Temporal Striding (ATS) — degradation without coverage loss.** Most systems suppress sub-models under adverse weather, silently dropping violation coverage. ATS instead processes every 5th frame for tracked vehicles. A vehicle in ROI for 30–60 frames still yields 6–12 classification samples — enough for 3-frame consensus within the 40 ms budget. Coverage is preserved; latency is not sacrificed. *(Layer 2A)*

2. **Computationally honest — ≤28 ms standard, ≤38 ms adverse peak.** The standard-condition cycle time (≤28 ms) is a design target on a specific device (Jetson Orin NX 16 GB), with the 3–5 ms IPC margin included. It holds when no adverse-weather preprocessing is active and the per-Track-ID gating in Layer 2A prevents both triggered sub-models from running concurrently on the same frame. The adverse peak (≤38 ms, with ATS striding classification to every 5th frame) is a *sustained-throughput* target held via inter-frame CUDA stream pipelining between preprocessing and YOLO inference; under worst-case adverse conditions the per-frame latency is higher and is reported separately (see compute-budget note). Sustained cadence p95 ≤ 38 ms is the go/no-go gate for Phase 2 — the system does not advance without meeting it. *(Layer 2A)*

3. **Self-healing spatial geometry — absent from every documented Indian ANPR deployment.** A 1° camera tilt silently shifts a virtual stop-line polygon by 1–3 metres on the ground plane. The Auto-Calibration Heartbeat detects this via SIFT/ORB homography every 15 minutes and re-projects all ROIs via $p' = H^{-1}p$ without halting enforcement. If landmark match rate drops below 50%, the junction enters *Calibration-Uncertain* mode and all detections route to human review. *(Layer 1A)*

4. **Character-level probability fusion — readable plates from mud.** Standard multi-frame OCR votes on the highest-confidence full string. Sentinel-BLR stores per-character softmax vectors across 5 frames, sums them element-wise, and takes argmax per position. A character visible in 2 frames but occluded in 3 still contributes probability mass. Levenshtein distance is an outlier filter, not the voting mechanism — a distinction that matters for partially-damaged plates. *(Layer 5)*

5. **Policy engine as YAML — a governance property, not just a technical one.** Enforcement rules live in a configuration file, not a model weight. BTP can add a diversion polygon, raise the OCR floor, or suspend auto-fining for a violation under legal review in under one minute, without engineer involvement or model redeployment. Auditability and updateability are structural guarantees. *(Layer 4A)*

6. **Closed-loop strategic intelligence — the components that can adapt, adapt.** The recency-decay formula (0.95^{days}) shifts junction urgency weight rapidly from synthetic priors to live data. Weekly LoRA fine-tuning on reviewer accept/reject decisions adapts the *MobileNetV3 safety classifier* specifically to Bengaluru lighting, plate fonts, and traffic geometry. To be precise about scope: this nightly loop updates the classification head, not the YOLOv10-N detector or the PaddleOCR model — those are re-trained on a slower Phase-boundary cadence once enough reviewer-labelled detection/OCR corrections accumulate, since gradient updates to detection and OCR backbones carry higher regression risk and run on cloud GPU, not the edge device. The claim is therefore "the adaptable component improves weekly," not "every model improves nightly." *(Layer 8)*

7. Layer-by-Layer Technical Specification

The Compute Story: How 40 ms Is Met

Stage	Activation Condition	Typical	Peak (adverse)
CLAHE (contrast)	Always active	∼5 ms	∼5 ms
Multi-Scale Retinex	Always active	∼8 ms	∼8 ms
DCP de-hazing	IMD API rain/fog OR ASTRAM Event Dataset waterlogging	—	+12 ms
Wiener de-blur	Laplacian variance blur detection	—	+10 ms
YOLOv10-N TensorRT INT8	Always	4 ms	7 ms
ByteTrack	Always	2 ms	4 ms

Stage	Activation Condition	Typical	Peak (adverse)
MobileNetV3 (triggered)	On 2-wheeler/car bounding box	3 ms	ATS stride
Triple-riding module (triggered)	On 2-wheeler bounding box (separate trigger from MobileNetV3)	*not concurrent*	ATS stride
IPC + pipeline overhead	Always	3–5 ms	3–5 ms
End-to-end frame cycle		**≤28 ms**	**≤38 ms (ATS)**

Compute budget assumptions — stated explicitly.

The ≤28 ms standard-condition figure holds when: (a) no adverse-weather preprocessing (DCP/Wiener) is active; and (b) at most one triggered classification sub-model runs per frame — either MobileNetV3 *or* the triple-riding module, not both simultaneously (per-Track-ID gating in Layer 2A prevents concurrent activation on the same vehicle). Always-active stages sum to 22 ms (CLAHE + Retinex + YOLO + ByteTrack) plus up to 3 ms for a single sub-model, plus 3 ms IPC = 28 ms.

The ≤38 ms adverse-peak figure is a *sustained-throughput* target — one fully processed frame emitted every ≤38 ms — and not a single-frame end-to-end latency claim. This distinction is stated explicitly because it is where naive compute budgets quietly cheat. With all four preprocessing stages active, the sequential preprocessing chain alone (CLAHE + Retinex + DCP + Wiener ≈ 35 ms) sits on the critical path of any individual frame, so worst-case *per-frame* latency is necessarily higher than 38 ms; preprocessing for frame N cannot be hidden behind YOLO inference of the same frame, because YOLO consumes that frame's preprocessed output. What CUDA stream parallelism buys is *inter-frame* pipelining: while frame N is preprocessed on one stream, YOLO inference for frame $N-1$ runs on a second stream, so the pipeline's output cadence approaches max(preprocessing, inference) rather than their sum. This holds throughput at the 38 ms cadence while adding roughly one pipeline stage of latency to each frame. Adequate violation coverage survives this added latency because ATS already strides classification to every 5th frame and a vehicle in ROI for 30–60 frames still yields 6–12 samples. Phase 1 reports the two numbers separately — sustained inter-frame cadence and per-frame p95 latency — and the go/no-go gate is the sustained cadence, with per-frame latency disclosed rather than asserted to also be ≤38 ms.

What Phase 1 will validate: Sustained inter-frame cadence p95 ≤ 38 ms over a 24-hour load test on the target device, with per-frame end-to-end latency reported alongside it (not assumed equal). The system does not advance to Phase 2 if the cadence target is missed; the ATS striding window will be tuned (e.g. every 4th frame instead of every 5th) to close any gap.

Under extreme conditions (all 4 preprocessing stages active), Layer 2A applies ATS: classification sub-models process every 5th frame for tracked vehicles.

Layer 3 — Five Parallel Violation Paths

- **Path A — Helmet (2-wheeler) / Seatbelt (4-wheeler):** two MobileNetV3 binary heads with separate triggers — the helmet head fires on a two-wheeler bounding box, the seatbelt head on a car/cab bounding box (a single binary classifier cannot serve two distinct tasks on two vehicle classes). Helmet is initialised from IDD + ACFR + Roboflow Helmet weights with 3-frame consensus. Seatbelt detection through a windshield is materially harder — glare, tint, reflection, and shallow viewing angle routinely defeat it — so in Phase 1 seatbelt is *human-review-only and never auto-fines*, regardless of confidence; it is promoted to auto-fine eligibility only if Phase 2 windshield-specific training data demonstrates ≥95% precision on a held-out night/glare set. This is the most honest scoping of the seven categories: the hardest one does not get to issue automated fines until it has earned it.

- **Path B — Triple Riding (Hybrid):**
  - *Primary* (crowd index > 0.35, dominant at Bengaluru peak-hour junctions): Volumetric Bounding-Box Profiling — flag raised if vertical/horizontal mass distribution exceeds $\Delta \geq 1.2\times$ single-rider baseline. 3-frame consensus.
  - *Enhancement* (crowd index ≤ 0.35): Chassis-Person Centroid Association with velocity coupling. MediaPipe Pose Lite extracts torso centroids within the expanded two-wheeler BB; only centroids within 15° of vehicle heading and 0.5 m/s tolerance counted.
  - Disagreement between both active paths routes to human review.
- **Path C — Wrong-Side:** Safe-Flow Vectors generated via SIFT/ORB + OSM road geometry. Deviation threshold: 25° (configurable per junction). 2-frame sustain. Minimum velocity filter eliminates stationary U-turns. *Risk flagged:* OSM directionality in Indian cities is frequently incomplete or stale, so OSM is used to *initialise* flow vectors only — they are corrected against observed dominant traffic flow per junction during Phase 1 commissioning, and any junction whose one-way status is contested is held at human-review-only until verified. The base directionality signal is not trusted blind.
- **Path D — Stop-Line/Red-Light:** YOLO lamp-colour classifier (primary, robust to sodium-vapour lighting and rain); BTP Signal API is a Phase 2 aspirational enhancement. Violation fires on simultaneous centroid-in-ROI and classifier-RED.
- **Path E — Illegal Parking:** Stationary-threshold filter; 180-second duration (configurable). No-parking ROI from manual annotation, BMTC GTFS bus-stop polygons, and Police Violations Dataset parking-concentration zones. Velocity floor 0.05 m/s excludes red-light stops.

Layer 4 — Confidence Gate and Escalation Router

Violation Type	Auto-Flag Threshold	Human Review Band	Discard Below
Helmet / seatbelt	≥ 88% / 3 frames	60–87%	< 60%
Triple riding	≥ 92% / 3 frames	65–91%	< 65%
Wrong-side driving	≥ 87% / 2 frames	60–86%	< 60%
Stop-line / red-light	≥ 90% / 3 frames	60–89%	< 60%
Illegal parking	≥ 85% / sustained	55–84%	< 55%

Triple riding carries the highest threshold (92%) because density-related false-positive risk is elevated. Illegal parking carries the lowest (85%) because sustained duration provides structurally stronger evidence.

Layer 4A — Enforcement Policy Engine

Violation	Condition	Enforcement Action
Triple riding	Any confidence	Human approval required — never auto-fine
Helmet (2-wheeler)	$\geq$ 88%, 3 frames, OCR confirmed	Auto-fine
Helmet (2-wheeler)	$\geq$ 88%, 3 frames, OCR low-confidence	Human approval required
Seatbelt (4-wheeler, Phase 1)	Any confidence	Human approval required — never auto-fine until Phase 2 windshield validation
Wrong-side driving	Near active diversion polygon	Human review only
Stop-line / red-light	$\geq$ 90%, 3 frames, signal confirmed	Auto-fine
Illegal parking (1st offence)	$\geq$ 85%, sustained 180–300 s, OCR confirmed	Warning only; auto-fine on 2nd offence within 30 days
Any violation	OCR confidence $<$ 90%	Human approval — never auto-fine
Any violation	Calibration-Uncertain junction	Human review only — no auto-fine

Layer 5 — Temporal Character-Level Probability Fusion

1. **Per-character softmax storage:** Probability vectors stored for each character position independently across up to 5 frames (chosen for maximum plate visibility).

2. **Position alignment (the step that makes summation valid):** Element-wise summation across frames is only meaningful if character *positions* are registered first. Frames at different distances/angles can segment a different number of characters (a spurious 10th glyph, a dropped leading character), and naively summing position i of frame A with position i of frame B would then corrupt the result. Frames are aligned by anchoring on the KA-regex plate template (state code + RTO + series + number block) and right/left-justifying segmented characters to that template; frames whose segmentation length cannot be reconciled to the template within edit distance 1 are dropped at this stage, not summed.

3. **Matrix summation:** The aligned per-position matrices are summed element-wise across all frames. Characters clearly visible in 2 frames but partially occluded in 3 still contribute probability mass.

4. **Argmax per position:** Final plate string is argmax of the summed matrix — highest-probability character at each position across all frames.

5. **Levenshtein outlier filter:** Applied *before* fusion to discard corrupt frames (e.g. mud splash). This is not the voting mechanism.

6. **Majority-vote confirmation:** If at least 3/5 frames agree within edit distance 1, the string is confirmed. Otherwise: LOW CONFIDENCE flag routes to human review.

Correction pairs for Indian number-plate font: 0/O, 1/I, 8/B, S/5. KA-regex gate validates Karnataka plate format.

Layer 6 — Evidence Packet and ERS

Each confirmed violation generates a tamper-evident packet: annotated frame + GradCAM overlay + UTC timestamp + GPS + junction ID + plate text + violation type + vehicle class + SHA-256 hash + **ERS (0–100)**.

$$\mathrm{ERS} = 0.30\,C_{\mathrm{det}} + 0.25\,C_{\mathrm{ocr}} + 0.20\,R_{\mathrm{consensus}} + 0.15\,H_{\mathrm{cal}} + 0.10\,Q_{\mathrm{pre}} \tag{2}$$

The SHA-256 hash covers the annotated frame bytes; any modification *after* hashing is detectable. To make this legally meaningful rather than merely technically true, the hash is anchored at generation time: each packet hash is written to an append-only, timestamped evidence log (RFC 3161 trusted-timestamp token, or a periodic Merkle-root commit), so the record proves both integrity and the time of capture and cannot be back-dated or silently regenerated. A bare hash with no trusted time anchor proves only that bytes did not change after an unspecified moment — insufficient for a contested challan — which is why the timestamp authority is part of the evidence design, not an afterthought.

Active Learning Feedback Loop: Every human accept/reject in the 60–92% confidence band is logged as a labelled training example. Weekly LoRA fine-tuning (Sunday 02:00) updates MobileNetV3 with low-rank decomposition to prevent catastrophic forgetting. Rising reviewer-override rate on the last 500 labelled examples triggers ahead-of-schedule fine-tuning.

8. Strengths and Remaining Limitations

Strengths

Capability	Detail
Full violation coverage	All seven violation categories in a single integrated pipeline
Computationally feasible	TensorRT INT8 + ATS + trigger gating + async queues → ≤28 ms standard / ≤38 ms adverse-peak design target on Jetson Orin NX; Phase 1 measured validation required
Self-healing geometry	SIFT/ORB landmark homography heartbeat detects and re-projects spatial ROIs without halting enforcement
Robust triple-riding	Volumetric BB Profiling is the primary method; MediaPipe Pose Lite activates as precision enhancement; both require 3-frame consensus
Temporal OCR fusion	Character-level softmax probability summed across 5 frames; Levenshtein outlier filtering; argmax per position
Repeat offender tracking	Cross-junction plate-violation index; REPEAT OFFENDER flag; MV Act 2019 §194C/§194D enhanced repeat-offence penalty authority
GradCAM explainability	Saliency overlay on every flagged frame; feeds reviewer acceleration and active learning loop
Evidence Reliability Score	Single composite 0–100 ERS on every packet; one number replaces five fields for officers and courts

Capability	Detail
Policy engine as config	Enforcement rules updated without redeploying any model; effective in < 1 minute
Hybrid Kafka payload	JSON metadata + compressed ROI JPEG (<50 KB); cloud human review without raw-frame streaming bandwidth
Legal grounding	MV Act 2019 §129 (helmet), §128/§194C (triple riding — rider limit and penalty), §184(e) (wrong-side/dangerous driving), §194D (enhanced penalty on repeat offence); enforcement policy traceable to statute

Remaining Limitations and Mitigations

Limitation	Current Mitigation	Future Path
Kannada-script plates	Temporal probability fusion + LOW CONFIDENCE flag routes to human review	Fine-tune PaddleOCR on Kannada-script plate dataset (Phase 2)
Night-time triple riding (IR-only)	IR-mode flag triggers luminance-based head-count heuristic; routed to human review	Dedicated IR-trained classifier
Safety violation data scarcity	Uniform priors (provided datasets contain 2 helmet and 0 triple-riding records across 298,450 rows; explicitly flagged in all outputs); IDD/ACFR/Roboflow Helmet pre-trained weights as classification foundation; <20% weight after 30 days live data	Replace uniform priors entirely with Phase 1 local data within 30 days
Single-camera blind spots	Deployment panel flags junctions needing multi-angle coverage via OSM geometry	Phase 3 multi-camera rollout

9. Deployment Pathway

Phase	Scope	Gate Criterion
Phase 1	10 high-urgency junctions (Police Violations Dataset top-10 parking urgency)	Go/no-go: Layer 2 mAP@0.5 $\geq$ 0.85 AND Layer 4 Precision $\geq$ 95% on Phase 1 data

Phase	Scope	Gate Criterion
Phase 2	50 junctions, selected by live Sentinel-BLR urgency scores	Synthetic priors replaced by locally grounded live data within 30 days of Phase 1
Phase 3	City-wide rollout with multi-angle coverage recommendations from Layer 9	Infrastructure cost grows linearly or sub-linearly with junction count

10. Gap Analysis — Why Manual Enforcement Fails

Bengaluru's road network spans approximately 14,000 km. The BTP's CCTV network generates far more footage than any human team can review in real time. Manual enforcement has three structural failure modes:

- **Scale** — more camera-hours than human officer capacity.
- **Environmental** — degraded performance in low-light, rain, and haze.
- **Strategic** — deployment based on officer intuition rather than violation density evidence.

Sentinel-BLR closes all three.

What Earlier Versions Did Not Get Right (v3.0 Red-Team Critique)

Critique (v3.0)	v6.0 Structural Response
Full pipeline at 25 fps is infeasible	Trigger-based inference + TensorRT INT8 + ATS + async queues → $\leq$28 ms standard / $\leq$38 ms adverse peak (CUDA stream parallelism required for peak). Phase 1 p95 validation target $\leq$ 38 ms.
MediaPipe Pose fails at high crowd density	Architecture inverted: Volumetric BB Profiling is now the primary path at high-density junctions (crowd index $>$ 0.35). MediaPipe activates as precision enhancement only.
OCR defeated by mud, damage, Kannada script	Temporal Character-Level Probability Fusion across 5 frames; softmax sum + argmax; Levenshtein outlier filter
Calibration drift when cameras shift	Auto-Calibration Heartbeat via SIFT/ORB every 15 min; automatic ROI re-projection
Police Violations Dataset has near-zero safety violation examples	Uniform priors (2 helmet, 0 triple-riding records across 298,450 rows; explicitly flagged) + pre-trained IDD/ACFR/Roboflow Helmet weights as classification foundation; not used for urgency scoring
Kafka raw frame streaming is network-prohibitive	Hybrid Evidence Payload: JSON metadata + Base64 ROI JPEG crop (<50 KB). Never raw video.

11. Operational Safeguards

Risk	Mitigation	Layer
Triple-riding false positives	Human approval always required regardless of confidence level	4A
OCR failure (dirty / Kannada plates)	No OCR confirmation = no auto-fine; routes to human review automatically	4A + 5
Camera occlusion during calibration	Scene occupancy monitor delays heartbeat if vehicle coverage $> 50\%$; retries in 60 s; 45-min max drift-window cap → Calibration-Uncertain mode	1A
Wrong-side context staleness	Officer draws diversion polygon via dashboard with expiry timestamp; 4A enforces no auto-fine within polygon	4A + 9
Review queue overload	Adaptive confidence floor rises when queue depth exceeds configurable limit, reducing inflow until queue clears	4
Privacy (non-violating road users)	Faces blurred before storage (Digital Personal Data Protection Act, 2023 compliance — notified Nov 2025, provisions phasing in through May 2027; BTP's CCTV-scale biometric processing plausibly qualifies it as a Significant Data Fiduciary under §10, which is tracked as a Phase 2 compliance review item); only violation-confirmed frames at full resolution; raw frames deleted within 24 hours	6 + config
Plate misread + repeat-offender index	Only HIGH CONFIDENCE reads (OCR $\geq 90\%$, majority-vote consensus) are indexed; *escalation* of a penalty under repeat-offender status additionally requires either two independent high-confidence reads of the same plate across separate events or human confirmation before the enhanced fine is applied — because a single misread that triggers escalation harms an innocent owner more than an ordinary false positive	5A

12. Data Integration Strategy

Source	Records	Used For	Explicitly NOT Used For
Police Violations Dataset (Nov 2023–Apr 2024)	298,450	Parking spatial heatmap; peak-hour patterns by vehicle class; junction deployment priority ranking	Safety violation urgency scoring — the dataset holds only 2 helmet and 8 seatbelt records across 298,450 rows; 97.3% of all violation tokens are parking offences. This is a parking dataset, not a safety dataset
ASTRAM Event Dataset (Nov 2023–Apr 2024)	8,173	Waterlogging events (458): pre-schedule DCP de-hazing. Accident events (365): 1.15× urgency multiplier	—
Uniform priors (provided datasets contain 2 helmet and 0 triple-riding records; explicitly labelled as uninformative in all outputs)	N/A	Safety-violation weight initialisation via IDD/ACFR/Roboflow Helmet pre-trained weights; <20% weight after 30 days live data; never presented as observed enforcement data	Never used for urgency scoring or auto-fine logic
IMD Weather API	Real-time	DCP de-hazing activation (15-min polling)	—
YOLO lamp-colour classifier; BTP Signal API (Phase 2)	Real-time	Signal state for stop-line detection	—
BMTC GTFS	Static	Bus-stop no-parking zone polygons	—
OSM / Maps Roads API	Static	Safe-Flow Vector initialisation; junction geometry for blind-spot analysis	—

Phase 1 junction set — derived directly from the provided data, not assumed. Ranking the 298,450 Police Violations records by junction shows enforcement load is heavily concentrated: the ten highest-volume named junctions account for a large share of all junction-attributed records (150,570 of 298,450 records carry a real junction tag). The Phase 1 deployment set is taken from the top of this ranking: Safina Plaza (BTP051, 15,449 records), KR Market (BTP082, 11,538), Elite Junction (BTP040, 10,718), Sagar Theatre (BTP044, 10,549), Central Street (BTP211, 5,388), Subbanna (BTP058, 5,189), Modi Bridge (BTP027, 4,584), Hosahalli Metro (BTP020, 4,101), Anand Rao (BTP057, 3,935), and NR Road/SP Road (BTP080, 3,681). Activity peaks in the morning enforcement window (08:00–11:00 IST), which sets the load profile the ≤38 ms cadence target is sized against. These are real rows in the supplied dataset, so the deployment plan is auditable end-to-end rather than a round-number guess.

13. Technical Stack

Component	Technology	Rationale
Object detection	YOLOv10-N, TensorRT INT8	∼4 ms inference; 38.5 $AP_{50–95}$ on MS-COCO at 1.84 ms on T4 TensorRT FP16 [1]; INT8 calibrated on 500 Bengaluru traffic frames
Multi-object tracking	ByteTrack (edge-local)	80.3 MOTA, 77.3 IDF1 on MOT17 at 30 FPS [2]; no raw frame transmission; Track IDs managed on-device; only metadata and ROI payloads upstream
Safety classification	MobileNetV3 (trigger-gated)	Binary classification; never runs without two-wheeler trigger; initialized from IDD + ACFR + Roboflow Helmet weights
Pose / centroid	MediaPipe Pose Lite (density-gated)	Volumetric BB Profiling primary at high-density junctions; MediaPipe as precision enhancement at low density
Preprocessing	OpenCV 4.x, CUDA build	CLAHE, Multi-Scale Retinex, DCP, Wiener; GPU-accelerated; conditionally activated
Deep learning	PyTorch 2.x	Training, fine-tuning, and LoRA active learning. **LoRA fine-tuning runs on cloud GPU (EC2 G5 / GCP T4), not on Jetson** — the Jetson's 100 TOPS NVDLA is optimised for INT8 inference, not gradient computation. Updated weights are pushed to Jetson via the existing Kafka connection each Sunday at 02:00.
License plate OCR	PaddleOCR PP-OCRv4	76.8% baseline accuracy on ICDAR 2015; degrades to 60.7% avg under rain/fog augmentation [3]; Sentinel-BLR's 5-frame fusion targets recovery of degraded-condition accuracy to ≥80%; Levenshtein outlier filter
Calibration	SIFT/ORB + homography (OpenCV)	15-min drift detection; H^{-1} for automatic ROI re-projection
Message broker	Apache Kafka 3.x	Hybrid Evidence Payload; partitioned by junction ID; no raw video
API layer	Python FastAPI	Async; WebSocket for live dashboard

Component	Technology	Rationale
Time-series DB	PostgreSQL + TimescaleDB (PG 16 + TS 2.x)	High-write time-series; temporal SQL; repeat-offender index
Batch analytics	DuckDB 0.10x	298k-record Police Violations Dataset queries in < 1 s
Dashboard	Streamlit + Folium + Plotly	GIS-capable; 8 panels; calibration health; public compliance feed
Containerisation	Docker + docker-compose	Full-stack deployment; reproducible environment
Edge hardware	NVIDIA Jetson Orin NX 16/64 GB	100 TOPS; full TensorRT pipeline at 25 fps; ATS + trigger-gating validated against TOPS specification
Cloud fallback	AWS EC2 G5 / GCP T4	On-demand GPU offload; Hybrid Kafka payload reduces egress bandwidth

14. Performance Evaluation Framework

Layer / Output	Metric(s)	Evaluation Method	Target
Layer 2 — detection	mAP@0.5, per-class precision/recall	Held-out test from Phase 1 footage, stratified by lighting/weather	mAP@0.5 ≥ 0.85; $\leq 10\%$ mAP drop in adverse conditions (YOLOv10n shows $\sim$8–12% mAP drop under fog/rain on DAWN dataset [4])
Layer 3/4 — auto-fine violations	Precision, Recall, F1	Confusion matrix vs. officer-adjudicated outcomes	Detector Precision $\geq 95\%$; **ERS-gated auto-fine Precision** $\geq 99\%$; Recall $\geq 80\%$
Layer 3/4 — triple riding	Recall, F1 (precision secondary)	Same protocol, scored separately	Recall $\geq 90\%$; all flags human-reviewed
Layer 5 — OCR	Plate-level accuracy, CER	Stratified holdout: clean / dirty / Kannada plates; before and after temporal fusion	$\geq 95\%$ clean; $\geq 80\%$ degraded (post-fusion)

Layer / Output	Metric(s)	Evaluation Method	Target
Layer 2A — latency	Sustained cadence p95 + per-frame p50/p95	24-hour sustained load test on Jetson Orin NX	Sustained cadence p95 $\leq$ 38 ms (gate); per-frame latency reported (Phase 1 measured; 3–5 ms IPC margin included)
System-wide — scalability	Junctions per edge node; Kafka + TimescaleDB throughput	Simulated load: 10 $\rightarrow$ 50 $\rightarrow$ city-wide	Linear or sub-linear infrastructure cost growth

Precision and recall targets are deliberately asymmetric. Auto-fine outcomes require high precision (false positive = wrongly issued fine). Triple riding — always human-reviewed — requires high recall (the only failure mode is missing a real violation). The two-tier precision target is deliberate and is set against the deployed baseline: Hyderabad's reported wrong-challan episode amounted to roughly 0.2% of all challans issued in the same period [8] (about 26,000 out of ~1.17 crore), yet that sub-1% error rate was still large enough in absolute terms to damage public trust. A 95% *detector* precision is therefore not an acceptable *auto-fine* bar — it would imply a far higher wrongful-fine rate than the system this proposal critiques. The ERS gate exists precisely to lift the auto-fine subset to ≥99% precision, so that Sentinel-BLR's automated-fine error rate is structurally lower, not higher, than the system it replaces; everything the gate is not confident about routes to human review rather than to a fine. The evaluation framework validates the policy choices already made in Layer 4A.

15. Cost and Legal Feasibility

Most submissions for this problem stop at the model. Two questions a deploying agency like BTP will ask before procurement — *what does this cost per junction*, and *does the evidence actually hold up under the statute it claims* — are addressed honestly below, with real-world figures rather than placeholder estimates.

Per-Junction Hardware Cost (Indicative)

Item	Indicative Cost (per junction)
Jetson Orin NX 16 GB module	~$600–970 (INR 50,000–80,000) retail per unit; NVIDIA does not publish fixed pricing, bulk/distributor quotes typically run lower. Carrier board, enclosure, NVMe storage, and power supply are separate line items not included in module price.

(continued on next page)

Item	Indicative Cost (per junction)
Camera + housing (existing BTP CCTV reused where possible)	Marginal cost *only where* the existing RTSP/ONVIF feed is already OCR-grade. This is the load-bearing assumption in the cost story and must not be overstated: much surveillance CCTV is sited for wide-area monitoring, not plate capture (angle, focal length, resolution-at-distance), so a fraction of junctions will need a dedicated plate camera — a junction-specific capex item Phase 1 surveys before committing. Full new-install cost is out of scope for this estimate.
Networking / cloud fallback (AWS EC2 G5 or GCP T4, on-demand)	Pay-per-use; Hybrid Evidence Payload design (JSON + <50 KB JPEG) keeps egress cost low relative to raw-frame streaming alternatives.

Phase 1 (10 junctions) is therefore primarily an edge-compute capital cost, in the range of a few lakh rupees for hardware alone — not the multi-crore infrastructure commitment a full IoT camera rollout would otherwise imply, because Sentinel-BLR is designed to sit on top of existing CCTV feeds rather than replace them. This is a deliberately conservative estimate: it excludes integration labour, BTP-side IT onboarding, and ongoing cloud spend, which are junction- and contract-specific and cannot be honestly quantified without procurement-level detail.

Legal Grounding — Corrected and Verified

Earlier drafts of this proposal cited Motor Vehicles Act sections that, on verification against the statutory text, did not say what was claimed. This version corrects them:

- **Helmet non-compliance** — §129 (headgear requirement), penalty under §194D.
- **Triple riding** — §128 (rider limit: driver + one pillion only), penalty under §194C.
- **Wrong-side / dangerous driving** — §184(e), *not* §185 (which is drunken driving, a distinct offence Sentinel-BLR does not address).
- **Repeat offences** — enhanced penalties are not governed by a single generic clause; they are built into the penalty sub-sections of §194C/§194D themselves (escalated fine + disqualification on subsequent conviction). §210A, occasionally miscited for this purpose, is in fact a State Government power to apply a 1×–10× fine multiplier by vehicle class — a separate, real mechanism BTP could use to calibrate fines for commercial vs. private two-wheelers, which Sentinel-BLR's Layer 4A policy engine could expose as a configurable parameter in a future iteration.

Data Protection Compliance — Corrected

Earlier drafts referenced a "PDPB 2023." This is not a real instrument: the Personal Data Protection Bill (PDPB) was an earlier draft withdrawn in 2022. The actual governing law is the **Digital Personal Data Protection Act, 2023 (DPDP Act)**, notified 13 November 2025, with provisions phasing in through 13 May 2027. Given the scale of biometric (facial) data a city-wide CCTV deployment would process, BTP's implementation plausibly meets the threshold for classification as a **Significant Data Fiduciary** under §10 of the Act, which carries additional obligations (data protection officer, periodic audits, data protection impact assessments). Sentinel-BLR's existing design choices — face-blurring before storage, 24-hour raw-frame deletion, violation-confirmed-only full-resolution retention — align directly with DPDP principles of data minimisation and storage limitation, but a formal compliance review against the phased commencement schedule is flagged here as a Phase 1 prerequisite rather than asserted as already satisfied.

16. Conclusion

Sentinel-BLR is an evidence-quality system designed to survive contact with the real world — the real Bengaluru, with cameras that shift in the wind, monsoon rain that obscures plates, markets where three bikes ride side by side at two metres per second, and an enforcement environment where an automated fine issued on low-reliability evidence is a legal liability, not a civic service.

The architectural choices documented here — Adaptive Temporal Striding, crowd-density-gated Volumetric Bounding-Box Profiling, Temporal Character-Level Probability Fusion, Hybrid Evidence Payloads, auto-calibration heartbeat, confidence-separated enforcement policy, legal grounding in MV Act 2019, and a graduated enforcement tier that builds public trust before applying fines — each exist because a specific, credible failure mode demanded them.

The result is a system that is technically proficient, operationally honest, legally defensible, and incrementally deployable — starting at ten high-urgency junctions identified from Police Violations Dataset analysis, improving every night it runs, and scaling to city-wide deployment without architectural replacement.

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[2] Zhang, Y., Sun, P., Jiang, Y., Yu, D., Weng, F., Yuan, Z., Luo, P., Liu, W., & Wang, X. (2022). *ByteTrack: Multi-Object Tracking by Associating Every Detection Box.* In *Proceedings of ECCV 2022*, Lecture Notes in Computer Science, vol. 13682. Springer. [80.3 MOTA, 77.3 IDF1, 63.1 HOTA on MOT17 test set at 30 FPS.]

[3] PaddlePaddle Authors. (2023). *PP-OCRv4: A Practical Ultra-Lightweight OCR System.* PaddleOCR v2.7 release, Baidu Inc. [PP-OCRv4-mobile: +4.5% Chinese scene, +10% English scene accuracy over PP-OCRv3. Adverse-weather accuracy on ICDAR 2015: 76.8% baseline, 60.7% average under rain/fog augmentation (Mazzeo et al., 2025).]

[4] Mazzeo, P.L., et al. (2025). *Robustness Analysis of YOLO and Faster R-CNN for Object Detection in Realistic Weather Scenarios with Noise Augmentation. Scientific Reports*, 15, 3211. Nature Publishing Group. [YOLOv10n evaluated on DAWN dataset (fog, rain, snow, sandstorm): mAP drop of 8–12% under adverse conditions; motivates Sentinel-BLR's CLAHE + DCP preprocessing pipeline and ATS degradation strategy.]

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[6] Deccan Herald. *Contactless Fines: Traffic Cops Back on Bengaluru Roads.* [M. N. Anucheth, Joint Commissioner of Police (Traffic), Bengaluru, on the relaunch of physical enforcement drives: ITMS does not detect violations such as wrong-side driving and footpath riding/parking, and registering such cases from surveillance footage remains a manual process.]

[7] Deccan Chronicle (via PressReader). *ANPR to Check Traffic Violations*. [Hyderabad's Automatic Number Plate Recognition system, launched under the city's Integrated and Intelligent Traffic Management System, detecting signal jumping, overspeeding, helmet, seatbelt, and mobile-phone-use violations.]

[8] Hyderabad Mail. *Smart Traffic System Turns Sour: 26,000 Wrong Challans Issued in Hyderabad*. [Cyberabad, Hyderabad, and Rachakonda traffic commissionerates issued over 26,000 wrong challans in the first ten months of 2025; officials attributed the majority to the ANPR violation-detection and plate-reading systems failing to sync.]

[9] ThePrint, reporting a Parliament Standing Committee action-taken report. *Parliament Panel Urges AI-Led Traffic Management, Global Best Practices for Delhi Roads*. [Delhi Police has installed 334 Red Light Violation Detection / Over-Speed Violation Detection cameras, with 328 more under procurement, as part of an ongoing AI-based Integrated Traffic Management System project.]

— END OF SUBMISSION —

Sentinel-BLR v6.0 — Evidence-Quality Traffic Enforcement Framework